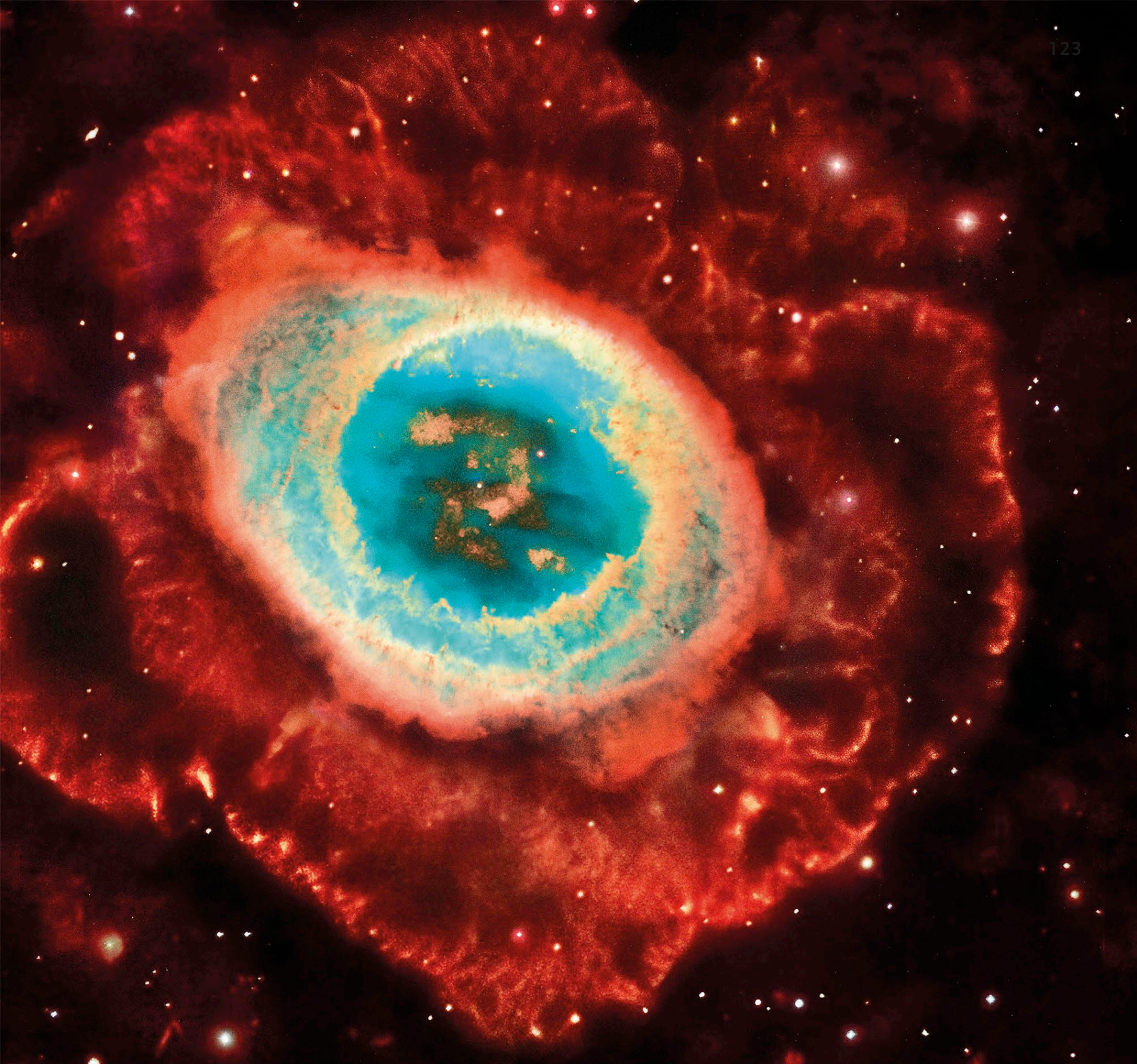




3 Captured on film

This image shows the Ring Nebula as captured on film in 1973, nearly 200 years after it was discovered. The nebula was discovered in 1779, independently by French astronomers Antoine Darquier de Pellepoix and Charles Messier. The picture was taken using the 13 ft (4 m) telescope at Kitt Peak National Observatory, Arizona. Astronomical images were not routinely recorded digitally for another decade.

4 False-color details

Taken three decades after picture 3 (above), this view of the Ring Nebula was also recorded at Kitt Peak National Observatory, this time using the 11½ ft (3.5-meter) telescope. It combines separate images taken through different colored filters. A red one highlights hydrogen and nitrogen, and a green one isolates oxygen. The use of these filters helps bring out greater detail in the nebula's shells.

5 Shape and structure

The nebula's shape is more complicated than appears at first glance. Its overall shape is similar to that of a barrel, appearing round because the "barrel" is positioned end-on to us (see p.120). Its blue center is in the shape of a football and this protrudes out of opposite sides of an orange-red doughnut ring of material. Dark knots of dense gas are embedded on the inner edge of the rim.

6 Dying star

Data from telescopes based in space and on the ground combine to give a complete picture of the Ring Nebula. The tiny white dot at the center is a white dwarf, the central remains of the star that blew off the surrounding material thousands of years ago. The nebula's ring shape is just under one light-year across, and the whole nebula is getting larger, expanding at 43,000 miles (69,000 km) per hour.